

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
%pip install xlrd
```

```
Collecting xlrd
```

```
  Downloading xlrd-2.0.2-py2.py3-none-any.whl.metadata (3.5 kB)
```

```
Downloading xlrd-2.0.2-py2.py3-none-any.whl (96 kB)
```

```
Installing collected packages: xlrd
```

```
Successfully installed xlrd-2.0.2
```

```
Note: you may need to restart the kernel to use updated packages.
```

```
[notice] A new release of pip is available: 25.1.1 -> 26.2.1
```

```
[notice] To update, run: C:\Users\pc\AppData\Local\Programs\Python\Python313\python.exe -m pip install --upgrade pip
```

```
gdp = pd.read_excel("gdp_data.xls", skiprows=3)
```

```
gdp.head()
```

	Country Name	Country Code	\
0	Aruba	ABW	
1	Africa Eastern and Southern	AFE	
2	Afghanistan	AFG	
3	Africa Western and Central	AFW	
4	Angola	AGO	

	Indicator Name	Indicator Code	1960
1961	\		
0	Total natural resources rents (% of GDP)	NY.GDP.TOTL.RT.ZS	NaN
NaN			
1	Total natural resources rents (% of GDP)	NY.GDP.TOTL.RT.ZS	NaN
NaN			
2	Total natural resources rents (% of GDP)	NY.GDP.TOTL.RT.ZS	NaN
NaN			
3	Total natural resources rents (% of GDP)	NY.GDP.TOTL.RT.ZS	NaN
NaN			
4	Total natural resources rents (% of GDP)	NY.GDP.TOTL.RT.ZS	NaN
NaN			

	1962	1963	1964	1965	...	2016	2017	2018
2019	\							
0	NaN	NaN	NaN	NaN	...	0.003344	0.002827	0.001885
0.001680								
1	NaN	NaN	NaN	NaN	...	7.241550	7.921979	7.572946
6.743400								
2	NaN	NaN	NaN	NaN	...	0.617410	0.702982	0.855264
0.682288								
3	NaN	NaN	NaN	NaN	...	6.360027	8.878331	10.002526

```
8.626806
4    NaN    NaN    NaN    NaN    ...    20.885120    25.950330    32.045406
28.897859
```

```

      2020      2021      2022      2023      2024      2025
0    0.003037    0.002066    NaN    NaN    NaN    NaN
1    6.150652   10.330017    NaN    NaN    NaN    NaN
2    0.305978    0.428525    NaN    NaN    NaN    NaN
3    5.917264    9.734380    NaN    NaN    NaN    NaN
4   20.247665   29.967435    NaN    NaN    NaN    NaN
```

```
[5 rows x 70 columns]
```

```
gdp = pd.read_excel("gdp_data.xls", skiprows=3)
```

```
resources = pd.read_excel("natural_resources.xls", skiprows=3)
```

```
print("GDP Dataset:")
```

```
print(gdp.head())
```

```
print("\nNatural Resources Dataset:")
```

```
print(resources.head())
```

```
GDP Dataset:
```

```

              Country Name Country Code \
0                      Aruba          ABW
1  Africa Eastern and Southern          AFE
2                      Afghanistan          AFG
3  Africa Western and Central          AFW
4                      Angola          AGO
```

```

              Indicator Name      Indicator Code      1960
1961 \
0  Total natural resources rents (% of GDP)  NY.GDP.TOTL.RT.ZS    NaN
NaN
1  Total natural resources rents (% of GDP)  NY.GDP.TOTL.RT.ZS    NaN
NaN
2  Total natural resources rents (% of GDP)  NY.GDP.TOTL.RT.ZS    NaN
NaN
3  Total natural resources rents (% of GDP)  NY.GDP.TOTL.RT.ZS    NaN
NaN
4  Total natural resources rents (% of GDP)  NY.GDP.TOTL.RT.ZS    NaN
NaN
```

```

      1962      1963      1964      1965      ...      2016      2017      2018
2019 \
0    NaN    NaN    NaN    NaN    ...    0.003344    0.002827    0.001885
0.001680
1    NaN    NaN    NaN    NaN    ...    7.241550    7.921979    7.572946
6.743400
2    NaN    NaN    NaN    NaN    ...    0.617410    0.702982    0.855264
```

```

0.682288
3  NaN  NaN  NaN  NaN  ...  6.360027  8.878331  10.002526
8.626806
4  NaN  NaN  NaN  NaN  ...  20.885120  25.950330  32.045406
28.897859

```

```

      2020      2021  2022  2023  2024  2025
0  0.003037  0.002066  NaN  NaN  NaN  NaN
1  6.150652  10.330017  NaN  NaN  NaN  NaN
2  0.305978  0.428525  NaN  NaN  NaN  NaN
3  5.917264  9.734380  NaN  NaN  NaN  NaN
4  20.247665  29.967435  NaN  NaN  NaN  NaN

```

[5 rows x 70 columns]

Natural Resources Dataset:

	Country Name	Country Code	Indicator Name \
0	Aruba	ABW	GDP (current US\$)
1	Africa Eastern and Southern	AFE	GDP (current US\$)
2	Afghanistan	AFG	GDP (current US\$)
3	Africa Western and Central	AFW	GDP (current US\$)
4	Angola	AGO	GDP (current US\$)

	Indicator Code	1960	1961	1962
1963 \				
0	NY.GDP.MKTP.CD	NaN	NaN	NaN
1	NY.GDP.MKTP.CD	2.420573e+10	2.495893e+10	2.707328e+10
2	NY.GDP.MKTP.CD	NaN	NaN	NaN
3	NY.GDP.MKTP.CD	1.190481e+10	1.270772e+10	1.363059e+10
4	NY.GDP.MKTP.CD	NaN	NaN	NaN

	1964	1965	...	2016	2017
2018 \					
0	NaN	NaN	...	2.983637e+09	3.092428e+09
1	3.027960e+10	3.380623e+10	...	8.326614e+11	9.792406e+11
2	NaN	NaN	...	1.811657e+10	1.875346e+10
3	1.580356e+10	1.692088e+10	...	6.999795e+11	6.942064e+11
4	NaN	NaN	...	6.077005e+10	8.562960e+10

	2019	2020	2021	2022
--	------	------	------	------

2023	\				
0	3.347562e+09	2.461056e+09	2.889305e+09	3.326671e+09	3.746573e+09
1	1.019354e+12	9.385461e+11	1.113060e+12	1.226461e+12	1.179122e+12
2	1.879944e+10	1.995593e+10	1.426000e+10	1.449724e+10	1.715223e+10
3	1.026528e+12	9.636882e+11	1.026670e+12	1.063748e+12	9.384807e+11
4	8.119381e+10	5.851203e+10	7.828392e+10	1.282340e+11	1.060423e+11

	2024	2025
0	4.167588e+09	NaN
1	1.252564e+12	1.358694e+12
2	1.777851e+10	NaN
3	7.373670e+11	8.525270e+11
4	1.030805e+11	1.221749e+11

[5 rows x 70 columns]

```
gdp = pd.read_excel("gdp_data.xls", skiprows=3)
```

```
resources = pd.read_excel("natural_resources.xls", skiprows=3)
```

```
print(pd.read_excel("gdp_data.xls", skiprows=3)["Indicator  
Name"].iloc[0])
```

```
print(pd.read_excel("natural_resources.xls", skiprows=3)["Indicator  
Name"].iloc[0])
```

Total natural resources rents (% of GDP)
GDP (current US\$)

```
resources = pd.read_excel("gdp_data.xls", skiprows=3)
```

```
gdp = pd.read_excel("natural_resources.xls", skiprows=3)
```

Natural Resources dataset

```
resources = pd.read_excel("gdp_data.xls", skiprows=3)
```

GDP dataset

```
gdp = pd.read_excel("natural_resources.xls", skiprows=3)
```

```
countries = [
    "United States",
    "China",
    "India",
    "Saudi Arabia",
    "Singapore"
]
```

```
gdp_5 = gdp[gdp["Country Name"].isin(countries)]
resources_5 = resources[resources["Country Name"].isin(countries)]
```

```
print("GDP:")
display(gdp_5[["Country Name", "2021"]])
```

```
print("\nNatural Resources:")
display(resources_5[["Country Name", "2021"]])
```

GDP:

	Country Name	2021
40	China	1.820170e+13
108	India	3.167271e+12
204	Saudi Arabia	9.826611e+11
207	Singapore	4.411109e+11
250	United States	2.372564e+13

Natural Resources:

	Country Name	2021
40	China	1.707565
108	India	3.159345
204	Saudi Arabia	25.565242
207	Singapore	0.000181
250	United States	1.279944

Keep only the required columns

```
gdp_final = gdp_5[["Country Name", "2021"]].copy()
resources_final = resources_5[["Country Name", "2021"]].copy()
```

Rename columns

```
gdp_final.columns = ["Country", "GDP_2021"]
resources_final.columns = ["Country", "Natural_Resources_%"]
```

Merge both datasets

```
country_data = pd.merge(
    gdp_final,
    resources_final,
    on="Country"
)
```

Show final table

```
country_data
```

	Country	GDP_2021	Natural_Resources_%
0	China	1.820170e+13	1.707565
1	India	3.167271e+12	3.159345
2	Saudi Arabia	9.826611e+11	25.565242

```

3      Singapore  4.411109e+11      0.000181
4  United States  2.372564e+13      1.279944

```

Normalize GDP and Natural Resources to a 0-100 scale

```

country_data["GDP_Score"] = (
    country_data["GDP_2021"] / country_data["GDP_2021"].max()
) * 100

```

```

country_data["Resource_Score"] = (
    country_data["Natural_Resources_%"] /
    country_data["Natural_Resources_%"].max()
) * 100

```

Create weighted Country Valuation Score

```

country_data["Country_Valuation_Score"] = (
    country_data["GDP_Score"] * 0.7 +
    country_data["Resource_Score"] * 0.3
)

```

Round the scores

```

country_data = country_data.round(2)

```

country_data

	Country	GDP_2021	Natural_Resources_%	GDP_Score	\
0	China	1.820170e+13	1.71	76.72	
1	India	3.167271e+12	3.16	13.35	
2	Saudi Arabia	9.826611e+11	25.57	4.14	
3	Singapore	4.411109e+11	0.00	1.86	
4	United States	2.372564e+13	1.28	100.00	

	Resource_Score	Country_Valuation_Score
0	6.68	55.71
1	12.36	13.05
2	100.00	32.90
3	0.00	1.30
4	5.01	71.50

Sort countries by valuation score

```

chart_data = country_data.sort_values(
    "Country_Valuation_Score",
    ascending=False
)

```

Create bar chart

```

plt.figure(figsize=(10, 6))

```

```

plt.bar(
    chart_data["Country"],
    chart_data["Country_Valuation_Score"]
)

```

```

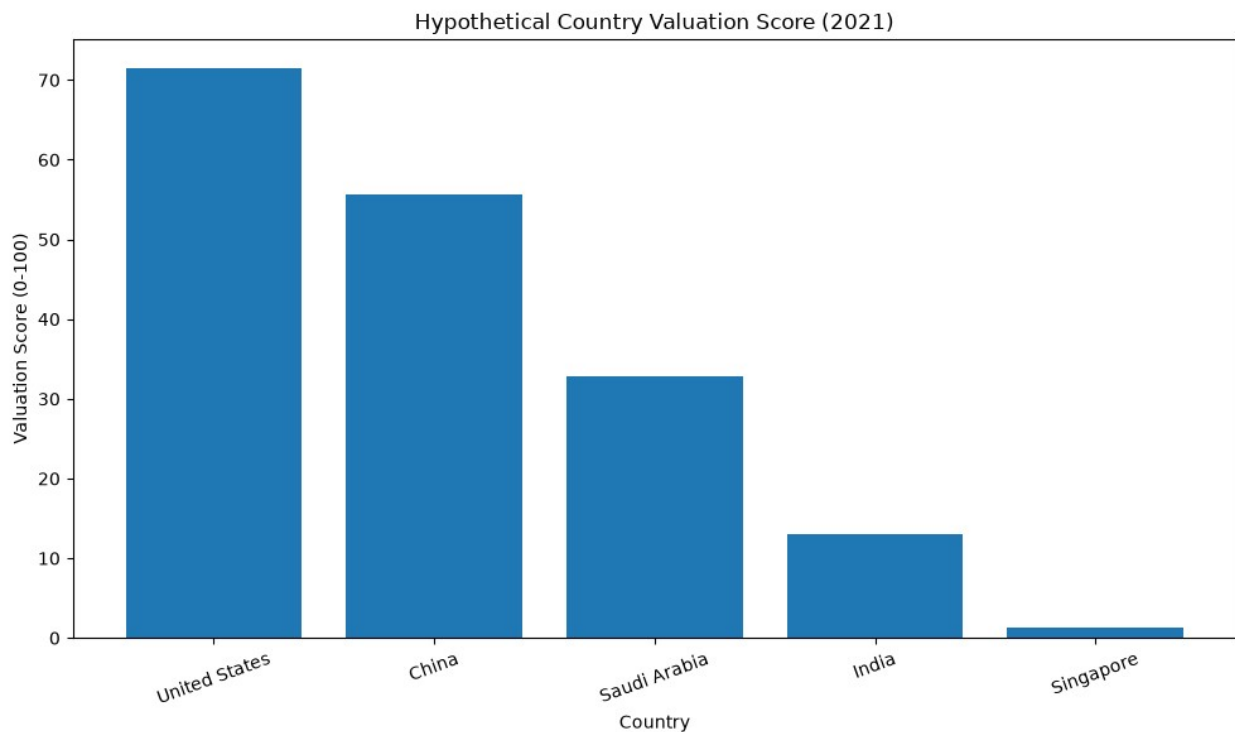
)

plt.title("Hypothetical Country Valuation Score (2021)")
plt.xlabel("Country")
plt.ylabel("Valuation Score (0-100)")

plt.xticks(rotation=20)
plt.tight_layout()

plt.show()

```



```

final_results = country_data[
    ["Country", "GDP_2021", "Natural_Resources_%",
     "GDP_Score", "Resource_Score", "Country_Valuation_Score"]
].sort_values("Country_Valuation_Score", ascending=False)

```

final_results

	Country	GDP_2021	Natural_Resources_%	GDP_Score \
4	United States	2.372564e+13	1.28	100.00
0	China	1.820170e+13	1.71	76.72
2	Saudi Arabia	9.826611e+11	25.57	4.14
1	India	3.167271e+12	3.16	13.35
3	Singapore	4.411109e+11	0.00	1.86
	Resource_Score	Country_Valuation_Score		
4	5.01	71.50		

0	6.68	55.71
2	100.00	32.90
1	12.36	13.05
3	0.00	1.30

This analysis values countries hypothetically as if they were assets using two factors: economic size and natural resources. GDP was given a weight of 70%, while natural resource rents were given a weight of 30%. Based on 2021 World Bank data, the United States achieved the highest valuation score of 71.50, followed by China at 55.71. Saudi Arabia ranked third because of its very high natural resource value. India and Singapore received lower scores under this specific framework. The analysis shows that a country's hypothetical value depends significantly on the factors and weights selected. A more advanced model could additionally include human capital, infrastructure, strategic importance, and country risk.